

The University of Texas at Austin

IN-CLASS WORK 9

Introduction to Financial Mathematics

October 01, 2026

EQUITY-LINKED PRODUCTS.

Problem 9.1.

An insurance company sells single premium deferred annuity contracts with return linked to a stock index. The time- t value of one unit of that stock index is denoted by $S(t)$. The contracts offer a minimum guaranteed return rate of g . At time 0, a single premium of amount π is paid by the policyholder, and $\pi \times y$ is deducted by the insurance company. Thus, at the contract maturity date, T , the insurance company will pay the policyholder

$$\pi \times (1 - y) \times \max[S(T)/S(0), (1 + g)^T].$$

You are given the following information:

- The contract will mature in one year.
- The minimum guaranteed rate of return, g , is 2%.
- Dividends are incorporated in the stock index. That is, the stock index is constructed with all stock dividends reinvested.
- $S(0) = 100$.
- The continuously compounded risk-free interest rate is 4%.
- The price of a one-year European call option, with strike price of 102, on the stock index is 12.40.

Determine y so that the insurance company does not make or lose money on this contract.

Solution. Set $K = S(0)(1 + g)^T = 100(1.02) = 102$. Then, the payoff of the variable annuity can be written as

$$\pi \frac{1 - y}{S(0)} \times \max[S(T), K].$$

Using the identity

$$\max[S(T), K] = K + \max[S(T) - K, 0],$$

the payoff of $\max[S(T), K]$ can be replicated by the following:

- a zero-coupon bond paying the redemption amount $K = 102$ at maturity $T = 1$, and
- one European call on the index with strike 102 and exercise date $T = 1$.

So the entire contract is replicated by $\pi(1 - y)/S(0)$ units of this portfolio. In our usual notation, its time-0 cost is

$$\frac{\pi(1 - y)}{S(0)} [Ke^{-rT} + V_{C(0)}] = \frac{\pi(1 - y)}{100} [102e^{-0.04} + 12.40].$$

Since $102e^{-0.04} \approx 98.00$,

$$\frac{\pi(1-y)}{100}(98.00 + 12.40) = \pi(1-y)(1.1040).$$

This would be the cost of exact hedging by replication. So, to break even, this cost must equal the premium π , which yields

$$(1-y)(1.1040) = 1 \implies 1-y = \frac{1}{1.1040} \approx 0.9058 \implies y \approx 9.42\%.$$